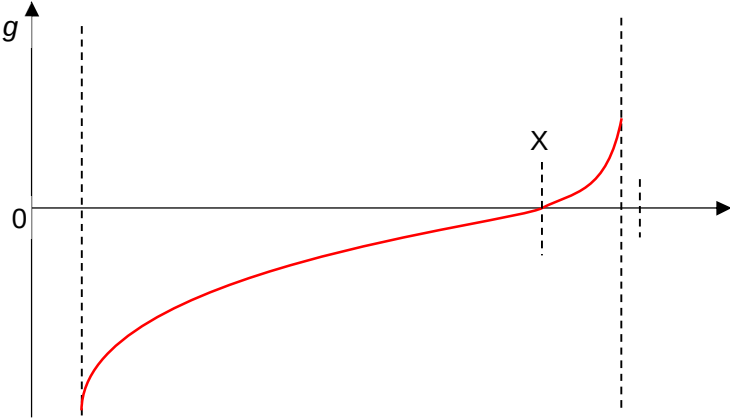


4(a)	gravitational force per unit mass experienced by a small test mass at that point in the field	B1
4(b)(i)	$\left. \begin{aligned} &\text{gravitational force provides centripetal force} \\ &\frac{GMm_i}{r^2} = m_i r \omega^2 \end{aligned} \right\}$ $\frac{GM}{r^3} = \left(\frac{2\pi}{T} \right)^2$ $T^2 = 4\pi^2 \frac{r^3}{GM}$	M1 A0
4(b)(ii)	$M = \frac{r^3}{G} \frac{4\pi^2}{T^2}$ $= \frac{(4.22 \times 10^5 \times 10^3)^3}{6.67 \times 10^{-11}} \frac{4\pi^2}{(1.77 \times 24 \times 60^2)^2}$ $= 1.90 \times 10^{27} \text{ kg}$	A1
4(c)	$\left. \begin{aligned} &g_j = g_i \\ &\frac{GM}{X^2} = \frac{Gm_i}{(r-X)^2} \end{aligned} \right\}$ $\frac{M}{m_i} = \frac{X^2}{(r-X)^2}$ $\frac{X}{r-X} = \sqrt{\frac{1.90 \times 10^{27}}{8.93 \times 10^{22}}}$ $= 146$	C1 A1

Qns	Answer	Marks
4(d)	 B1: shape, negative at LHS, positive at RHS B1: g at jupiter surface more magnitude than at Io surface B1: X is x-intercept much nearer to Io	
4(e)	hi speed electrons hitting metallic outside of spacecraft results in X-ray photons highly energetic so may damage onbaord equipment.	M1 A0
5(a)	force per unit length per unit current	B2